Nikolai Slavov

CONTACT INFORMATION

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ORCID ID: 0000-0003-2035-1820

Parallel Squared Technology Institute: parallelsq.org
Academic Laboratory: slavovlab.net
SCP Center: center.single-cell.net

EDUCATION

Princeton University, Princeton, USA

Ph.D. in Molecular and Quantitative Cell Biology **06.2010**

M.A. in Molecular Biology **02.2006**

Mentor: David Botstein

Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations

Massachusetts Institute of Technology (MIT), Cambridge, USA

B.S. in Biology **06.2004**

RESEARCH & ACADEMIC APPOINTMENTS

Parallel Squared Technology Institute (PTI), MA, USA

Founding Director **2023–present**

Northeastern University, Boston, USA

COE Distinguished Professor **2026–present**

Professor, Department of Bioengineering **2024–2026**

Faculty Fellow, Barnett Institute **2017–present**

Affiliated Faculty, Department of Chemistry and Chemical Biology **2016–present**

Affiliated Faculty, Department of Biology **2016–present**

Associate Professor, Department of Bioengineering **2021–2024**

Assistant Professor, Department of Bioengineering **2015–2021**

Harvard University, Cambridge, USA

Research Fellow, Departments of Systems Biology and Statistics **2014–2015**

Massachusetts Institute of Technology (MIT), Cambridge, USA

Research Fellow, van Oudenaarden laboratory, Departments of Biology and Physics **2011–2013**

AWARDS & FELLOWSHIPS

✓ Biemann Medal by the American Association for Mass Spectrometry **2026**

✓ COE Distinguished Professor **2026**

✓ HUPO Discovery in Proteomic Sciences Award **2022**

✓ College of Engineering Faculty Fellow **2022**

✓ Allen Distinguished Investigator ($1,500,000) **2020**

✓ Excellence in Mentoring Award **2018**

✓ NIH Director's New Innovator Award ($2,352,750) **2016**

✓ Broad Institute of Harvard and MIT SPARC Award **2014**

- ✓ Princeton University Dean's Award **2010**
- ✓ IRCSET Postgraduate Research Fellowship (72,000 €) **2007**
- ✓ Finalist in the Young European Entrepreneur Competition **2006**
- ✓ Princeton Graduate Fellowship ($ 45,000) **2004**
- ✓ MIT Undergraduate Fellowship ($ 125,000) **2001**
- ✓ Eureka Fellowship for Academic Excellence (10,000 €) **2000**
- ✓ Bronze Medal in the 31st International Chemistry Olympiad (*IChO*) **1999**
- ✓ National Diploma for Exceptional Achievements in Chemistry **1999**

Highlighted Peer Reviewed Publications

[OA] – Open Access article ✉ – Corresponding author Google Scholar Profile PubMed

- ✓ Tsour S, Machne R, Leduc A, Widmer S, Koo E, Guez J, Karczewski K, **Slavov N**✉ (2026)
  Alternate RNA decoding results in stable and abundant proteins in mammals
  Nature, 656, 506–515 DOI: 10.1038/s41586-026-10678-2

- ✓ Iwamoto-Stohl LK, Petelski AA, ... **Slavov N**✉, Zernicka-Goetz (2025)
  Fertilization triggers early proteomic symmetry breaking in mammalian embryos
  Cell, Vol. 188, 26, 7428 - 7444.e21, DOI: 10.1016/j.cell.2025.11.006 **[OA]**

- ✓ Huffman RG, Leduc A, Wichmann C, di Gioia M, Borriello F, Specht H,..., **Slavov N**✉ (2023)
  Prioritized mass spectrometry increases the depth, sensitivity and data completeness of single-cell proteomics
  Nature Methods, **20**, 714-722 DOI: 10.1038/s41592-023-01830-1 **[OA]**

- ✓ Derks J, Leduc A, Wallmann G, Huffman G, Willetts M, Khan S, Specht H,..., **Slavov N**✉ (2022)
  Increasing the throughput of sensitive proteomics by plexDIA
  Nature Biotechnology, DOI: 10.1038/s41587-022-01389-w

- ✓ Leduc A, Huffman RG, Cantlon J, Khan S, **Slavov N**✉ (2022)
  Exploring functional protein covariation across single cells using nPOP
  Genome Biology, **23**, 261 DOI: 10.1186/s13059-022-02817-5 **[OA]**

- ✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D,... & **Slavov N**✉ (2022)
  Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
  Genome Biology, 22, 50 DOI: 10.1186/s13059-021-02267-5 **[OA]**

✓ Budnik B, Levy E, Harmange G & **Slavov N** ✉ (2018)
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: 10.1186/s13059-018-1547-5 **[OA]**

✓ **Slavov N** ✉, Macinskas J, Caudy A, Botstein D (2011)
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 **[OA]**

Articles invited for revisions

✓ Derks J., McDonnell K., Wamsley N., Stewart P., Yeh M., Specht H., **Slavov N** ✉
Increasing mass spectrometry throughput using time-encoded sample multiplexing, *bioRxiv* DOI: 10.1101/2025.05.22.655515 ..... Invited revision at *Nature biotech*

✓ Specht H., Yeh M., Sipe S., Barnes-Seeman D., Adamo M., McDonnell K,... **Slavov N** ✉
PSMtags improve peptide sequencing and throughput in sensitive proteomics, *bioRxiv*, DOI: 10.1101/2025.05.22.655509 ..... Invited revision at *Nature methods*

Selected Peer Reviewed Publications

2026

✓ McDonnell K., Wamsley N., Derks J., Sipe S., Yeh M., Specht H., **Slavov N** ✉
JMod: Joint modeling of mass spectra for empowering multiplexed DIA proteomics, *Nature methods (in press)* DOI: 10.1101/2025.05.22.655512

✓ Leduc A, Zheng S, Saxena P, **Slavov N** ✉
Protein degradation and growth dependent dilution substantially shape mammalian proteomes
Molecular Systems Biology (in press) DOI: 10.1101/2025.02.10.637566

✓ Derks J, Jonson T, Leduc A, Khan S, Khoury L, Rafiee M, **Slavov N** ✉ (2024) Single-nucleus proteomics identifies regulators of protein transport
Genome Biology (in press) DOI: 10.1101/2024.06.17.599449

✓ **Slavov N** ✉
Developing Single-Cell Proteomics to Open New Perspectives on Biology
Journal of the American Society for Mass Spectrometry, 2026 DOI: 10.1021/jasms.6c00362

✓ Tsour S, Machne R, Leduc A, Widmer S, Koo E, Guez J, Karczewski K, **Slavov N** ✉
Alternate RNA decoding results in stable and abundant proteins in mammals
Nature, 656, 506–515 DOI: 10.1038/s41586-026-10678-2
↪ Multiomics Study Suggests Alternate RNA Translation More Pervasive Than Thought, highlight by *GenomeWeb*
↪ Analyzing alternate translation, highlight by *Nature Chemical Biology*
↪ When Variant Proteins Aren't Actually the Variant Ones, highlight by *Science*

✓ **Slavov N** ✉
Single-Cell Proteomic Technologies: Tools in the quest for principles
Annual Review of Biophysics, 55, 253–275 DOI: 10.1146/annurev-biophys-021424-011450
↪ How Single-Cell Proteomics Could Unlock the Rules That Govern Cells, highlight by *AZo Life Sciences*

2025 **2025**

✓ Iwamoto-Stohl LK, Petelski AA, ... **Slavov N** ✉, Zernicka-Goetz
Fertilization triggers early proteomic symmetry breaking in mammalian embryos
Cell, Vol. 188, 26, 7428 - 7444.e21, DOI: 10.1016/j.cell.2025.11.006
↪ Probing the proteome at cellular scale, highlight by *Nature*

✓ **Slavov N** ✉ *et al.*
Artificial intelligence in metabolic research
Nat Metab. 2025, DOI: 10.1038/s42255-025-01395-7

✓ Liu Y, ..., **Slavov N**, Murn J
Composition and RNA binding specificity of metazoan RNase MRP
Nucleic Acids Res 2025, 27;53(16):gkaf829 DOI: 10.1093/nar/gkaf829

✓ **Slavov N** ✉
Unlocking the Potential of Single-Cell Omics
J. Proteome Res. 2025, 24, 1481 – 1481 DOI: 10.1021/acs.jproteome.5c00197

✓ Leduc A, Xu Y, Shipkovenska G, Dou Z, **Slavov N** ✉
Limiting the impact of protein leakage in single-cell proteomics
Nature Communications, 5;16(1):4169 DOI: 10.1038/s41467-025-56736-7

✓ Rengarajan S, Derks J, Bellott D, **Slavov N**, Page DC
Post-transcriptional cross- and auto-regulation buffer expression of the human RNA helicases DDX3X and DDX3Y *Genome Research*, DOI: 10.1101/gr.279707.124

2024 **2024**

✓ Leduc A., Koury L., Cantlon J., Khan S., **Slavov N** ✉
Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP
Nature Protocols, DOI: 10.1038/s41596-024-01033-8

✓ SenNet Consortium, **Slavov N**, *et al.*
SenNet recommendations for detecting senescent cells in different tissues
Nat Rev Mol Cell Biol 25(12):1001-1023, DOI: 10.1038/s41580-024-00738-8

✓ Galatidou S, Petelski A, Pujol A, Lattes K,..., **Slavov N**, Barragan M
Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes
Molecular Human Reproduction, DOI: 10.1093/molehr/gaae023 **[OA]**

✓ Rogers ZJ, Colombani T, Khan S,..., **Slavov N**, Bencherif SA
Controlling Pericellular Oxygen Tension in Cell Culture Reveals Distinct Breast Cancer Responses to Low Oxygen Tensions
Adv Sci. e2402557, DOI: 10.1002/advs.202402557 **[OA]**

✓ Sipe SS, **Slavov N** ✉
Single-Cell Proteomics Accelerates toward Proteoforms
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.4c00290

✓ Khan S, Conover R, Asthagiri AR, **Slavov N**✉
Dynamics of single-cell protein covariation during epithelial-mesenchymal transition
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.4c00277 **[OA]**

2023 **2023**

✓ Wallmann G., Leduc A., **Slavov N**✉
Data-Driven Optimization of DIA Mass Spectrometry by DO-MS
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.3c00177 **[OA]**

✓ **Slavov N**✉
Single-cell proteomics: quantifying post-transcriptional regulation during development with mass-spectrometry
Development, 150 (13): dev201492, DOI: 10.1242/dev.201492

✓ van den Berg PR, Bérenger-Currias NMLP, Budnik B, **Slavov N**✉, Semrau S
Integration of a multi-omics stem cell differentiation dataset using a dynamical model
PLOS Genetics **19** (5) :1-23 DOI: 10.1371/journal.pgen.1010744 **[OA]**

✓ Huffman RG, Leduc A, Wichmann C, di Gioia M, Borriello F, Specht H, Derks J, Khan S,..., **Slavov N**✉
Prioritized mass spectrometry increases the depth, sensitivity and data completeness of single-cell proteomics
Nature Methods, **20**, 714-722 DOI: 10.1038/s41592-023-01830-1 **[OA]**

✓ Gatto L, Aebersold R, Cox J, Demichev V, Derks J, Emmott E, Franks AM, Ivanov AR,..., **Slavov N**✉
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
Nature Methods **20**, 375–386 DOI: 10.1038/s41592-023-01785-3 **[OA]**

✓ MacCoss MJ, Alfaro J, Faivre, D, Wu C, Wanunu M, **Slavov N**✉
Sampling proteomes by emerging single-molecule and mass-spectrometry methods
Nature Methods **20**, 339–346, DOI: 10.1038/s41592-023-01802-5

✓ **Slavov N**✉
Great Gains in Mass Spectrometry Data Interpretation
Journal of Proteome Research, **22** 3, 659 DOI: 10.1021/acs.jproteome.3c00099

✓ Derks J, **Slavov N**✉
Strategies for increasing the depth and throughput of protein analysis by plexDIA
Journal of Proteome Research **22** 697-705, DOI: 10.1021/acs.jproteome.2c00721 **[OA]**

2022 **2022**

✓ Leduc A, Huffman RG, Cantlon J, Khan S, **Slavov N**✉
Exploring functional protein covariation across single cells using nPOP
Genome Biology, **23**, 261 DOI: 10.1186/s13059-022-02817-5 **[OA]**
↪ Single-Cell Proteomics Bypasses Bottlenecks, Sways Skeptics, highlight by *GEN*

✓ Petelski AA, **Slavov N**, Specht H
Single-Cell Proteomics Preparation for Mass Spectrometry Analysis Using Freeze-Heat Lysis and an Isobaric Carrier
J. Vis. Exp., (190), DOI: 10.3791/63802

✓ Derks J, Leduc A, Wallmann G, Huffman G, Willetts M, Khan S, Specht H,..., **Slavov N** ✉
Increasing the throughput of sensitive proteomics by plexDIA
Nature Biotechnology, DOI: 10.1038/s41587-022-01389-w
↪ Sensitive protein analysis with plexDIA, highlight by *Nature Methods*
↪ Uncovering Proteomic Patterns One Cell at a Time, highlight by *GEN*

✓ Specht H, **Slavov N** ✉
Beyond Protein Sequence: Protein Isomerization in Alzheimer's Disease
Journal of Proteome Research 21 (2) :299-300, DOI: 10.1021/acs.jproteome.2c00016

✓ SenNet Consortium, **Slavov N**
NIH SenNet Consortium to map senescent cells throughout the human lifespan to understand physiological health
Nat aging 1090–1100 DOI: 10.1038/s43587-022-00326-5

✓ **Slavov N** ✉
Counting protein molecules for single-cell proteomics
Cell 185 (2) :232-234, DOI: 10.1016/j.cell.2021.12.013

✓ **Slavov N** ✉
Learning from natural variation across the proteomes of single cells
PLoS Biology 20(1): e3001512, DOI: 10.1371/journal.pbio.3001512 **[OA]**

✓ **Slavov N** ✉
Scaling up single-cell proteomics
Molecular & Cellular Proteomics 21(1), 100179, DOI: 10.1016/j.mcpro.2021.100179 **[OA]**

2021 **2021**

✓ Khoury L, **Slavov N** ✉
Comprehensive Identification of Regulatory Protein Networks
Journal of Proteome Research 20(11): 4913-4914, DOI: 10.1021/acs.jproteome.1c00813

✓ He L, *et al.*, **Slavov N**, Marneros A,
Global characterization of macrophage polarization mechanisms and identification of M2-type polarization inhibitors
Cell Reports 37(5):109955, DOI: 10.1016/j.celrep.2021.109955

✓ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N** ✉
Multiplexed single-cell proteomics using SCoPE2
Nature Protocols 16, 5398–5425, DOI: 10.1038/s41596-021-00616-z

✓ **Slavov N** ✉
Driving Single Cell Proteomics Forward with Innovation
Journal of Proteome Research 20, 11, 4915–4918, DOI: 10.1021/acs.jproteome.1c00639

✓ **Slavov N** ✉
Measuring Protein Shapes in Living Cells
Journal of Proteome Research DOI: 10.1021/acs.jproteome.1c00376

✓ **Slavov N**✉
Increasing proteomics throughput
Nature Biotechnology, 39, 809–810, DOI: 10.1038/s41587-021-00881-z

✓ **Slavov N**✉ *et al.*
Voices of biotech research: Single-cell proteomics
Nature Biotechnology, 39, 281–286, DOI: 10.1038/s41587-021-00847-1

✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, Serra M, Kharchenko P, Koller A & **Slavov N**✉
Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
Genome Biology, 22, 50 DOI: 10.1186/s13059-021-02267-5 **[OA]**
↪ Towards resolving proteomes in single cells, highlight by *Nature Methods*

2020 **2020**

✓ Specht H, **Slavov N**✉
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.0c00675

✓ Petelski AA **Slavov N**✉
Analyzing ribosome remodeling in health and disease
Proteomics, 20, (17) :2000039 DOI: 10.1002/pmic.202000039

✓ **Slavov N**✉
Single-cell protein analysis by mass-spectrometry
Current Opinion in Chemical Biology, 60, 1 - 9 DOI: 10.1016/j.cbpa.2020.04.018

✓ **Slavov N**✉
Unpicking the proteome in single cells
Science, 367 (6477) :512 - 513, DOI: 10.1126/science.aaz6695

2019 **2019**

✓ **Slavov N**✉ *et al.*
Voices in methods development: Single-cell proteomics
Nature Methods, 16, 945 - 951, DOI: 10.1038/s41592-019-0585-6

✓ Huffman RG, Chen AT, Specht H & **Slavov N**✉
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
Journal of Proteome Research, 18, 6, 2493 - 2500, DOI: 10.1021/acs.jproteome.9b00039
↪ Technology feature highlight by *Nature Methods*

✓ Chen AT, Franks A & **Slavov N**✉
DART-ID increases single-cell proteome coverage
PLoS Computational Biology, 5(7):e1007082 DOI: 10.1371/journal.pcbi.1007082 **[OA]**
↪ The single cell proteomics revolution, editorial highlight by *Bioanalysis Zone*

✓ Emmott EP, Jovanovic M & **Slavov N**✉
Approaches for studying ribosome specialization
Trends in Biochemical Sciences, 44(5):478 - 479, DOI: 10.1016/j.tibs.2019.01.008

✓ Emmott EP, Jovanovic M & **Slavov N**✉
Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, 44(2):95 - 109, DOI: 10.1016/j.tibs.2018.10.009

✓ Malioutov D., Chen T., Jaffe J., Airoldi E., Budnik B. & **Slavov N**✉
Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, 18(1):162 - 168, DOI: 10.1074/mcp.TIR118.000947

2018 **2018**

✓ Specht H. & **Slavov N**✉
Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: 10.1021/acs.jproteome.8b00257
↪ Innovations in Proteomics: The Drive to Single Cells, editorial highlight
↪ Through the Looking Glass of Single Cell Proteomics, highlight by *Technology Networks*

✓ Budnik B., Levy E., Harmange G. & **Slavov N**✉
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: 10.1186/s13059-018-1547-5 **[OA]**
↪ SCoPE-MS – We can finally do single cell proteomics!!!, highlight by *Proteomics News*
↪ Researchers Apply Mass Spec to Single-Cell Proteomics, highlight by *GenomeWeb*
↪ Single-cell proteomics highlight by *the NIH Director's office*
↪ Technology feature highlight by *Nature Methods*
↪ The single cell proteomics revolution, editorial highlight by *Bioanalysis Zone*
↪ Probing the proteome at cellular scale, highlight by *Nature*

✓ Levy E. & **Slavov N**✉
Single cell protein analysis for systems biology
Essays in Biochemistry, 62(4):595 - 605, DOI: 10.1042/EBC20180014

2017 **2017**

✓ Franks A., Airoldi E.M., **Slavov N**✉
Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: 10.1371/journal.pcbi.1005535 **[OA]**
↪ How Statistics Weakened mRNA's Predictive Power highlight by *The Scientist*

✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N**, Pasparakis M, Balachandran S, Kelliher M, Mecsas J, Degterev A.
Kinase Activities of RIPK1 and RIPK3 Can Direct IFN-β Synthesis Induced by Lipopolysaccharide.
The Journal of Immunology, 198(11):4435-4447, DOI: 10.4049/jimmunol.1601717

2016 **2016**

✓ Klionsky, *et al.*, **Slavov N**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, 12(1):1-222, DOI: 10.1080/15548627.2015.1100356

✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3):e0150605, DOI: 10.1371/journal.pone.0150605 **[OA]**

2015 **2015**

✓ **Slavov N**✉
Making the most of peer review
eLife, 4:e12708, DOI: 10.7554/eLife.12708 **[OA]**

✓ **Slavov N**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A.
Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: 10.1016/j.celrep.2015.09.056 **[OA]**
↪ All Ribosomes Are Created Equal. Really?, highlight by *Trends in Biochemical Sciences*
↪ New direction for tissue engineering and cancer therapeutics, highlight by *Science X*

✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N**, *et al.*
De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: 10.1016/j.cmet.2015.04.003

Started Slavov Lab at Northeastern University, August 2015

2014 **2014**

✓ Malioutov D.✉, **Slavov N**✉
Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 **[OA]**

✓ **Slavov N**✉, Budnik B., Schwab D., Airoldi E., van Oudenaarden A.✉
Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 **[OA]**

2013 **2013**

✓ **Slavov N**✉, Carey, J., Linse, S.✉
Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 **[OA]**

✓ **Slavov N**✉, Botstein D.✉
Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Deregulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 **[OA]**

2012 **2012**

✓ **Slavov N**✉, van Oudenaarden A.✉
How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 **[OA]**

✓ **Slavov N**✉, Airoldi E.M., van Oudenaarden A., Botstein D.✉
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 **[OA]**

2011 **2011**

✓ **Slavov N**✉, Macinskas J., Caudy A., Botstein D.✉
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 **[OA]**

✓ **Slavov N**✉, Botstein D.✉
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 **[OA]**

2010 **2010**

✓ **Slavov N**✉
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 **[OA]**

✓ **Slavov N**✉, Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University **[OA]**

✓ Silverman SJ., **Slavov N**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.✉
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 **[OA]**

2009 and before **2009 and before**

✓ **Slavov N**✉, Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079-4084 ("In This Issue" article) **[OA]**

✓ Tagkopoulos I.✉, **Slavov N**✉, Kung S.✉ (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5th IEEE Symposium on Bioinformatics and Bioengineering (BIBE'05).

Patents

- **Slavov N**, *et al.*, Mass tags for multiplexing. U.S. Provisional Patent Application No. 63/757,298.
- **Slavov N**, *et al.*, TimePLEX: Time-domain multiplexing for proteomics. U.S. Provisional Patent Application No. 63/791,304.
- **Slavov N**, *et al.*, IsoPLEX: Time-domain multiplexing with small molecule isomers. U.S. Provisional Patent Application No. 63/794,877.
- **Slavov N**, *et al.*, Single-cell proteomics analysis of embryos. U.S. Patent Application No. 19/249,653.
- **Slavov N**, *et al.*, Miniaturized Proteomic Sample Preparation. U.S. Patent Application No. 18/556,655.
- **Slavov N**, *et al.*, Methods of multiplexed data-independent acquisition for proteomics. U.S. Patent Application No. 17/806,468.
- **Slavov N**, *et al.*, Mass spectrometry technique for single cell proteomics, US Patent No. 11,719,703 issued on 8/8/2023.

Published Preprints

Google Scholar

▷ Kok O, Friedrich C, Altig K, Oakley D, Hyman BT, Collins M, **Slavov N** ✉ (2026)
Scalable proteomics with multiDIA resolves proteoforms in neurodegenerative disease
bioRxiv DOI: 10.64898/2026.10.16.756994

▷ Koury L, Leduc A, Khan S, Hagemann-Jensen M, Michaelsson J, Mold JE, **Slavov N** ✉ (2026)
Single-cell proteomics reveals cell-type-specific functional coordination in PBMCs
bioRxiv DOI: 10.64898/2026.09.11.751068

▷ Yue QX, Shen Y, Dai C, ..., **Slavov N**, ..., Perez-Riverol Y (2026)
quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets
Research Square DOI: 10.21203/rs.3.rs-10319687/v1

▷ Leduc A, Shipkovenska G, Xu Y, Franks A, **Slavov N** ✉ (2025)
The regulation of protein abundance across single cells in a mammalian tissue
bioRxiv DOI: 10.1101/2025.09.17.676955

▷ Sinn LR, Szyrwiel L, Grossmann J, Lau K, Faisst K, Qin D, Mutschler F, Khoury L, Leduc A, Ralser M, Coscia F, Selbach M, **Slavov N**, Nagaraj N, Steger M, Demichev V (2025)
Slice-PASEF: Maximising Ion Utilisation in LC-MS Proteomics
bioRxiv DOI: 10.1101/2022.10.31.514544

▷ **Slavov N** ✉ (2025)
Single-Cell Proteomic Technologies: Tools in the quest for principles
arXiv DOI: 10.48550/arXiv.2506.18198

▷ Collins M, Friedrich C, Elcheikhali M, Stewart P, Derks J, ..., **Slavov N**✉ (2025)
Ubiquitin-Proteasome System Dysregulation in Alzheimer's Disease Impacts Protein Abundance
bioRxiv DOI: 10.1101/2025.05.29.656728

▷ Derks, J., McDonnell, K., Wamsley, N., Stewart, P., Yeh, M., Specht, H., **Slavov N**✉ (2025)
Increasing mass spectrometry throughput using time-encoded sample multiplexing
bioRxiv DOI: 10.1101/2025.05.22.655515

▷ McDonnell, K., Wamsley, N., Derks, J., Sipe, S., Yeh, M., Specht, H., **Slavov N**✉ (2025)
JMod: Joint modeling of mass spectra for empowering multiplexed DIA proteomics
bioRxiv DOI: 10.1101/2025.05.22.655512

▷ Specht, H., Yeh, M., Sipe, S., Barnes-Seeman, D., Adamo, M., McDonnell, K., Agius, M. P., Friedrich, C., Pang, W. K., Huang, Y., Shiva Raju, K., Vuong, W., Lee, M. A., Yesilcimen, A., Pentelute, B. L., **Slavov N**✉ (2025)
PSMtags improve peptide sequencing and throughput in sensitive proteomics
bioRxiv, DOI: 10.1101/2025.05.22.655509

▷ Leduc A, **Slavov N**✉ (2025)
Protein degradation and growth dependent dilution substantially shape mammalian proteomes
bioRxiv DOI: 10.1101/2025.02.10.637566

▷ Khan S, Elcheikhali M, Leduc A, Huffman R, Derks J, Franks A, **Slavov N**✉ (2024)
Inferring post-transcriptional regulation within and across cell types in human testis
bioRxiv DOI: 10.1101/2024.10.08.617313

▷ Iwamoto-Stohl LK, Petelski AA, ... **Slavov N**, Zernicka-Goetz (2024)
Proteome asymmetry in mouse and human embryos before fate specification
bioRxiv DOI: 10.1101/2024.08.26.609777

▷ Tsour S, Machne R, Leduc A, Widmer S, **Slavov N**✉ (2024)
Alternate RNA decoding results in stable and abundant proteins in mammals
bioRxiv DOI: 10.1101/2024.08.26.609665
↪ When Variant Proteins Aren't Actually the Variant Ones, highlight by *Science*

▷ Leduc A, Xu Y, Shipkovenska G, Dou Z, **Slavov N**✉ (2024)
Limiting the impact of protein leakage in single-cell proteomics
bioRxiv DOI: 10.1101/2024.07.26.605378

▷ Derks J, Jonson T, Leduc A, Khan S, Khoury L, Rafiee M, **Slavov N**✉ (2024) Single-nucleus proteomics identifies regulators of protein transport
bioRxiv DOI: 10.1101/2024.06.17.599449

▷ Galatidou S, Petelski A, Pujol A, ..., **Slavov N**✉, Barragan M (2024)
Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes
bioRxiv DOI: 10.1101/2024.05.23.595547

▷ Khan S, Conover R, Asthagiri AR, **Slavov N**✉ (2023)
Dynamics of single-cell protein covariation during epithelial-mesenchymal transition
bioRxiv, DOI: 10.1101/2023.12.21.572913 **[OA]**

▷ Leduc A., Koury L., Cantlon J., **Slavov N**✉ (2023)
Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP
bioRxiv, DOI: 10.1101/2023.11.27.568927 **[OA]**

▷ Leduc A, Harens H, **Slavov N**✉ (2023)
Modeling and interpretation of single-cell proteogenomic data
arXiv DOI: 10.48550/arXiv.2308.07465

▷ Wallmann G., Leduc A., **Slavov N**✉ (2023)
Data-Driven Optimization of DIA Mass-Spectrometry by DO-MS
bioRxiv, DOI: 10.1101/2023.02.02.526809 **[OA]**

▷ Derks J, **Slavov N**✉ (2022)
Strategies for increasing the depth and throughput of protein analysis by plexDIA
bioRxiv, DOI: 10.1101/2022.11.05.515287 **[OA]**

▷ Gatto L, ..., **Slavov N**✉ (2022)
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
arXiv, DOI: 10.48550/arXiv.2207.10815 **[OA]**

▷ MacCoss MJ, Alfaro J, Wanunu M, Faivre DA, **Slavov N**✉ (2022)
Sampling proteomes by emerging single-molecule and mass-spectrometry methods
arXiv, DOI: 10.48550/arXiv.2208.00530 **[OA]**

▷ Huffman RG, Leduc A, ..., **Slavov N**✉ (2022)
Prioritized single-cell proteomics reveals molecular and functional polarization across primary macrophages
bioRxiv, DOI: 10.1101/2022.03.16.484655 **[OA]**

▷ Derks J, Leduc A, Huffman RG, Specht H, Ralser M, Demichev V, **Slavov N**✉ (2021)
Increasing the throughput of sensitive proteomics by plexDIA
bioRxiv, DOI: 10.1101/2021.11.03.467007 **[OA]**

▷ Leduc A, Huffman RG, **Slavov N**✉ (2021)
Droplet sample preparation for single-cell proteomics applied to the cell cycle
bioRxiv, DOI: 10.1101/2021.04.24.441211 **[OA]**

▷ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**✉ (2021)
Multiplexed single-cell proteomics using SCoPE2
bioRxiv, DOI: 10.1101/2021.03.12.435034 **[OA]**

▷ Specht H, **Slavov N**✉ (2020)
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
bioRxiv, DOI: 10.1101/2020.08.24.264994 **[OA]**

▷ Petelski AA, **Slavov N**✉ (2020)
Analyzing ribosome remodeling in health and disease
arXiv, arXiv:2007.05839 **[OA]**

▷ **Slavov N**✉ (2020)
Single-cell protein analysis by mass-spectrometry
arXiv, arXiv:2004.02069 **[OA]**

▷ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, *et al.* & **Slavov N**✉ (2019)
Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity *bioRxiv*, DOI: 10.1101/665307 **[OA]**
↪ Over 1,000 SINGLE CELL PROTEOMES!, highlight by *Proteomics News*
↪ Technology feature highlight by *Nature Methods*

▷ Specht H, Emmott E, Koller A, **Slavov N**✉ (2019)
High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
bioRxiv, DOI: 10.1101/665307 **[OA]**

▷ Huffman G, Specht H, Chen AT, **Slavov N**✉ (2019)
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
bioRxiv, DOI: 10.1101/512152 **[OA]**

▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N**✉ (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: 10.1101/399774 **[OA]**

▷ Chen A, Franks A, **Slavov N**✉ (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: 10.1101/399121 **[OA]**

▷ Levy E. & **Slavov N**✉ (2018)
Single cell protein analysis for systems biology
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26965v1 **[OA]**

▷ Emmott EP, Jovanovic M & **Slavov N**✉ (2018)
Ribosome stoichiometry: from form to function
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26991v1 **[OA]**

▷ Specht H. & **Slavov N**✉ (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26821v1 **[OA]**

▷ Malioutov D., Chen T., Jaffe J., Airoldi E., Carr S., Budnik B., **Slavov N**✉ (2017)
Quantifying homologous proteins and proteoforms
bioRxiv, DOI: 10.1101/168765 **[OA]**

▷ Berg P., Budnik B., **Slavov N**✉, Semrau S. (2017)
Dynamic post-transcriptional regulation during embryonic stem cell differentiation
bioRxiv, DOI: 10.1101/123497 **[OA]**

▷ Budnik B., Levy E., **Slavov N**✉ (2017)
Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
bioRxiv, DOI: 10.1101/102681 **[OA]**
↪ SCoPE-MS – We can finally do single cell proteomics!!!, highlight by *Proteomics News*
↪ Researchers Apply Mass Spec to Single-Cell Proteomics, highlight by *GenomeWeb*
↪ Interview, highlight by *Front Line Genomics*

▷ **Slavov N**✉ (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: 10.7287/peerj.preprints.1504v1 **[OA]**

▷ Franks A., Airoldi E.M., **Slavov N**✉ (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: 10.1101/020206 **[OA]**

▷ **Slavov N**✉, Botstein, D., Caudy A. (2014)
Extensive regulation of metabolism and growth during the cell division cycle
bioRxiv, DOI: 10.1101/005629 **[OA]**

▷ **Slavov N**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A. (2014)
Differential stoichiometry among core ribosomal proteins
bioRxiv, DOI: 10.1101/005553 **[OA]**

Opinions & Popular Science Publications

▷ **Slavov N**✉ (2025) Ion Mobility's Innovation Explosion
The Analytical Scientist, Link

▷ **Slavov N**✉ (2018) We Must Demand Evidence of Peer Review
The Scientist, Link

▷ **Slavov N**✉ (2017) What was rate limiting was the idea, not the technology
Front Line Genomics, Link

▷ **Slavov N**✉ (2015) Making the most of peer review
eLife, Link

▷ **Slavov N**✉ (2014) Accomplishments Over Accolades!
The Scientist, Link

▷ **Slavov N**✉ (2014) Discoveries lie hidden behind the façade of popular assumptions
Cell Reporter, Link

▷ **Slavov N**✉ (2013) The Best Projects Are Least Obvious
The Scientist, Link

▷ **Slavov N**✉ (2013) A Wonderful Christmas with Mysterious Oscillations
PubChase, Link

▷ **Slavov N**✉ (2012) The Mission of MIT
The MIT Tech, Link

Invited Keynote Talks

1. Single-cell proteomics: An exhilarating journey, Video
   Biemann Medal Award lecture at ASMS **June.02.2026**
2. Aging through the lens of single-cell proteomics
   Aging Research Summit, Albert Einstein College of Medicine **Feb.19.2026**
3. Increasing proteomics throughput by multiplexing in the mass and time domains
   Proteomics-Based Drug Discovery Summit **Feb.3.2026**
4. From Single-Cell Proteomics to Molecular Mechanisms
   6th Annual GCC Single Cell and Spatial Omics Symposium **Oct.22.2025**
5. From proteomics to molecular mechanisms
   BSPR Annual Meeting 2025 **Jul.07.2025**
6. Functional biological explorations with single-cell proteomics
   Single Cell and Spatial Biology in Precision Medicine **Dec.16.2024**
7. From proteomics to molecular mechanisms
   eXceed Symposia at ASMS 2024 **Jun.02.2024**
8. Uncovering the secrets of the human proteome, Video
   Barnett Institute 50th Anniversary Celebration **May.10.2024**
9. Advancing single-cell proteomics
   Advances in Proteomics Technologies (APT-2024) **Feb.19.2024**
10. Biological and technological frontiers for single-cell proteomics
    International single-cell mass spectrometry conference **Oct.5.2023**
11. Driving discoveries with single-cell proteomics
    Summer School on Computational Proteomics **June.27.2023**
12. Strategies for increasing the depth and throughput of single-cell proteomics
    ASMS Congress **June.4.2023**

13. Single-cell proteomics: An exhilarating journey
HUPO 2022 World Congress, Human Proteome Organization **Dec.7.2022**

14. Single-Cell Protein Analysis by Mass-Spectrometry, Video
ASMS Asilomar Conference on Single-Cell Mass Spectrometry **Oct.8.2022**

15. Scaling up single-cell analysis to the proteome
IMSC 2022 Conference **Aug.29.2022**

16. Exploring functional protein covariation across single cells
Next-Generation Protein Analysis and Detection **May.30.2022**

17. Learning from natural variation across the proteomes of single cells
XIV Annual Congress of the European Proteomic Association **Apr.4.2022**

18. Learning from natural variation across the proteomes of single cells
3rd IBSE International Symposium **Feb.3.2022**

19. Increasing the sensitivity, reliability, reproducibility and throughput of single-cell proteomics
HUPO Reconnect 2021 World Congress, Human Proteome Organization **Nov.15.2021**

20. Nano proteomic sample preparation for single-cell protein analysis
International Symposium on Microscale Separations and Bioanalysis **Jul.12.2021**

21. Droplet sample preparation for single-cell proteomics applied to the cell cycle
2021 AOHUPO, Human Proteome Organization **Jun.30.2021**

22. Computational opportunities for single-cell proteomics, Video
MaxQuant Summer School **Jun.25.2021**

23. Mapping the transcriptome and proteome of tissues in 3D at single-cell resolution
Plant Cell Atlas **Apr.08.2021**

24. Characterizing transcriptional and post-transcriptional regulation from single-cell variability
CELLINK Partnership Conference **Dec.11.2020**

25. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity (link)
The 2020 HUPO World Congress, Human Proteome Organization **Oct.20.2020**

26. Single-cell proteomics by mass-spectrometry (link)
Netherlands Proteomics Platform, Utrecht, Netherlands **Jan.17.2020**

27. Quantifying proteins in single cells at high-throughput
Single-cell Omics Symposium at Peking University **Oct.20.2019**

28. High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
European Single Cell Proteomics Conference, Institute of Molecular Pathology (IMP) **Aug.27.2019**

29. Quantifying proteins in single cells at high-throughput
The 28th International KOGO Annual Conference, South Korea **Sept.06.2019**

30. DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
MaxQuant Summer School, University of Wisconsin– Madison, Video **Jul.25.2019**

Invited Departmental Talks & Visits

1. Regulating protein abundance by protein degradation
   Dana Farber Cancer Institute, Harvard University **April.8.2026**
2. Secrets of the human proteome: From protein variation to biological functions
   University College Cork **Oct.10.2025**
3. Alternate RNA decoding results in stable and abundant proteins in mammals
   University of Notre Dame **March.2025**
4. Functional biological explorations with single-cell proteomics
   Harvard Medical School and Dana-Farber Cancer Institute **May.19.2025**
5. Uncovering secrets of the proteome: From protein variation to biological functions
   University of Georgia, Institute of Bioinformatics (IOB) **March.28.2025**
6. Functional biological explorations with single-cell proteomics
   Boston University, Medical School **Feb.12.2025**
7. Functional biological explorations with single-cell proteomics
   Braod Institute, Single cell seminar **Sept.17.2024**
8. Biological and technological frontiers for single-cell proteomics
   UMass Amherst **April.25.2024**
9. Biological and technological frontiers for single-cell proteomics
   MIT **Dec.05.2023**
10. Biological and technological frontiers for single-cell proteomics
    University of Chicago PME Distinguished Colloquium Series Seminar **Nov.17.2023**
11. Single-cell proteomics for chemical biology, Video
    Broad-MIT Seminar in Chemical Biology **Nov.06.2023**
12. Biological and technological frontiers for single-cell proteomics
    Yale University **Sept.26.2023**
13. Biological and technological frontiers for single-cell proteomics
    Purdue University **Oct.31.2023**
14. Biological and technological frontiers for single-cell proteomics
    University of Nebraska Medical Center **June.16.2023**
15. Exploring functional protein covariation across single cells
    Department of Biochemistry and Molecular Biology, UC Riverside **April.25.2023**
16. Exploring functional protein covariation across single cells
    Institute for Systems Biology **April.21.2023**
17. Exploring functional protein covariation across single cells
    Department of Pharmacology at UC San Diego **Feb.27.2023**
18. Driving biology with single-cell proteomics
    Eli Lilly **Nov.11.2022**
19. Functional protein covariation across single cells
    Systems Biology Division, NIGMS, NIH **Nov.10.2022**
20. Driving biology with single-cell proteomics: New data acquisition and interpretation methodologies, Video
    2022 IMaSS Single Cell Day **Oct.12.2022**
21. Investigating biological mechanisms with high-throughput single-cell proteomics
    Takeda **Sep.13.2022**

22. Exploring functional protein covariation across single cells, Video
GIGA University of Liege, Belgium **Sept.2.2022**

23. Exploring functional protein covariation across single cells
Laboratory of Receptor Biology and Gene Expression, NCI, NIH **Mar.9.2022**

24. Single-cell proteomic and transcriptomic analysis of macrophage polarization
NYU Pathology Department **Feb.14.2022**

25. Understanding biological systems: In search of direct causal mechanisms
D. E. Shaws **Jan.19.2022**

26. Single-cell proteomic and transcriptomic analysis of macrophage polarization
Oxford University and Charite' **Oct.20.2021**

27. Using Single-Cell Proteomics and Transcriptomics to Understand Gene Regulation
University of Texas Medical Branch **Sept.16.2021**

28. Single-cell proteomic and transcriptomic analysis of macrophage polarization
Department of Cell Biology, Duke University **Jan.14.2021**

29. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Peking University **Oct.11.2020**

30. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Department of Biochemistry, Boston University **Sept.14.2020**

31. Quantifying proteins in single cells at high-throughput
Center for Proteomics and Metabolomics, Leiden University **Jan.16.2020**

32. Quantifying proteins in single cells at high-throughput
Biotech Research and Innovation Center, University of Copenhagen **Jan.15.2020**

33. Coordination among metabolism and cell division
Boston Children Hospital & Harvard Medical School **Nov.14.2019**

34. Single-cell proteomics quantifies the emergence of macrophage heterogeneity
Department of Physiology, Johns Hopkins University School of Medicine **Nov.6.2019**

35. Understanding biological systems: In search of direct causal mechanisms (link) Video
Models, Inference & Algorithms Initiative (MIA), Broad Institute of MIT & Harvard **Apr.3.2019**

36. Understanding and controlling cell differentiation using single-cell mass-spectrometry (link)
Department of Systems Biology (Theory Lunch), Harvard Medical School **May.18.2018**

37. Principles of ribosome-mediated translational regulation
The Center for Engineering in Medicine, Massachusetts General Hospital **Mar.17.2017**

38. Principles of ribosome-mediated translational regulation
Systems Biology Seminar, Boston University **Oct.13.2016**

39. Ribosome-mediated translational regulation: Direct methods for quantifying the ribosome code
Center for Systems Biology, Columbia University **Apr.20.2016**

40. Coordinating metabolism and ribosome-mediated translational regulation
Department of Biology, New York University **Apr.19.2016**

41. Quantifying protein isoforms (link)
Models, Inference & Algorithms Initiative (MIA), Broad Institute **Sept.14.2015**

Invited Presentations for Funding Recommendations

- Scalable identification and quantification of protein modifications
  Power Workshop, MJFOX Fundation **Oct.28.2024**
- Scalable single-cell proteomics
  Allen Institute, Brain Initiative **March.20.2024**
- Opportunities for single-cell proteomics and metabolomics
  CZI **Oct.6.2022**
- Driving single-cell proteomics forward with innovation
  National Institute on Aging, NIH **Sept.16.2021**
- Current state of single-cell proteomics, standards, potentials and quantitative benchmarks
  Functional Proteomics, HubMap & Common Fund, NIH **Oct.8–9.2020**
- Single-cell proteomics
  Helmsley charitable trust, Gut cell atlas meeting, NYC **Oct.28.2019**
- Transformative opportunities for single-cell proteomics
  NHLBI Single-cell workshop, National Institutes of Health (NIH) **Oct.17.2019**

Invited Conference Talks

1. Increasing proteomics throughput by multiplexing in the mass and time domains (link)
   Biohub: Chicago BioEngineering Conference **Sept.17–18.2026**
2. Mechanobiology through the Lens of Proteomics (link)
   GRC: Multiscale Mechanochemistry and Mechanobiology **Jul.26–31.2026**
3. Single-cell proteomics: An exhilarating journey, Video
   Biemann Medal Award lecture at ASMS **June.02.2026**
4. From Transcripts to Proteins: Principles of Proteins Abundance Control across Single Cells in Mammalian Tissues (link)
   Gordon Research Conference: Single-Cell Genomics **May.10–15.2026**
5. Increasing proteomics throughput by multiplexing in the mass and time domains
   SLAS2026 International Conference **Feb.10.2026**
6. Scalible Single-cell proteomics
   Next-Generation Mass Spectrometry Summit on advanced 4D proteomics **Jan.29.2026**
7. Single-cell proteomics
   37th Annual Tandem Mass Spectrometry Workshop **Dec.4.2025**
8. Quantifying protein abundance regulation in single cells from complex tissues
   International Conference on Systems Biology, Dublin **Oct.8.2025**
9. From Protein Variation to Biological Functions
   Karger Medal Analytical Chemistry Celebration **March.10.2025**
10. Inferring molecular mechanisms from single-cell proteomics data
    NextGen Omics & Spatial Biology **March.27.2025**

11. Mechanobiology through the lens of proteomics
Mechanobiology Summit Conference **March.7.2025**
12. Joint protein and RNA analysis enables inferring post-transcriptional regulation in single cells from complex tissues
CZI Neuroscience Focus Group Meeting **Jan.12.2025**
13. Inferring molecular mechanisms from single-cell proteomics data
EMBL Cobference: Quantitative biology to molecular mechanisms **Nov.22.2024**
14. Accurate protein analysis of single cells and subcellular organelles reveals regulatory principles
Festival of Genomics **06.2024**
15. Single-nucleus proteomics identifies regulators of protein transport
ASMS **06.2024**
16. High-Throughput Single Cell Proteomics Analysis by Mass Spectrometry
NextGen Omics & Synthetic Biology US **03.2024**
17. High-throughput Single Cell Proteomics
SLAS2024 **Feb.6.2024**
18. Biological and technological frontiers for single-cell proteomics
CSHL Single-cell analysis **Nov.9.2023**
19. Single Cell Proteomics. Parallel Squared.
Allen Frontiers Group **April.21.2023**
20. High-throughput Single Cell Proteomics
SLAS2023 **March.1.2023**
21. Exploring functional protein covariation across single cells
The Scientist: Pushing Towards Single-Cell Resolution **Dec.12.2022**
22. Introduction to single-cell proteomics by mass-spectrometry, Recorded video
HUPO 2022 World Congress, Human Proteome Organization **Dec.4.2022**
23. Prioritized single-cell proteomics reveals molecular and functional polarization across primary macrophages
The 51st Annual Meeting of the Japanese Society for Immunology **Dec.6.2022**
24. Global analysis of post-transcriptional regulation with single-cell resolution
Tri-Omics, Oxford Global **Sept.28.2022**
25. Driving biology with single-cell proteomics: New data acquisition and interpretation methodology for sensitive protein analysis
Italian Mass Spectrometry Society (IMaSS) **Oct.12.2022**
26. Innovations in single-cell proteomics enable exploring functional protein covariation across thousands of single cells
International Symposium on MS in the Health and Life Sciences **Aug.14.2022**
27. Exploring functional protein covariation across single cells using parallel proteomics
Gordon Research Conference: Single-Cell Genomics **May.2.2022**
28. Single-cell proteomics
The Conference on Retroviruses and Opportunistic Infections **Feb.13.2022**
29. Droplet sample preparation for single-cell proteomics applied to the cell cycle
CZI Seed Network Webinar **Jul.20.2021**
30. Single-cell Proteomic and Transcriptomic Analysis of Macrophage Heterogeneity (link)
HUPO World Congress **Oct.20.2021**

31. Single-cell protein analysis by mass-spectrometry (link)
The future of single cell analysis, Earlham Institute **Feb.16.2020**

32. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity (link)
Applied Bioinformatics in Life Sciences **Feb.14.2020**

33. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity (link)
Cellular and Molecular Bioengineering Conference, BMES **Jan.3.2020**

34. Single-cell proteomics for studying regeneration and aging (link)
MDI Biological Symposium for Regeneration and Aging **Oct.9.2019**

35. Mapping the Transcriptome and Proteome of Human Testis in 3D
CZI Human Cell Atlas Seed Networks Meeting **Jul.30.2019**

36. Transformative opportunities for single cell proteomics
The Greater Boston Mass Spectrometry Discussion Group **Dec.13.2018**

37. Understanding and controlling cell differentiation based on comprehensive mass-spec quantification of proteins and signaling in single cells
UPENN Single Cell Biology Symposium, Video **Nov.13.2018**

38. Chair for the Single Cell Proteomics session of HUPO
Automated sample preparation for high-throughput single-cell proteomics **Oct.3.2018**

39. Using single-cell proteomics to engineer directed cell differentiation
Bioengineering & Translational Medicine Conference **Sept.27.2018**

40. Analyzing single cell proteomics data
ETH, Biognosys Discovery Proteomics Seminar **Jul.7.2018**

41. Mass–spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
CSHL Meeting: Single Cell Analyses **Nov.10.2017**

42. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Annual Single Cell Analysis USA Congress, Boston **Oct.23.2017**

43. Quantifying protein heterogeneity during the differentiation of single stem cells
Global CESI-MS Symposium **Oct.06.2017**

44. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Festival of Genomics, Boston **Oct.04.2017**

45. From differential transcription of ribosomal proteins to differential structure of ribosomes
Genetics, Genomics and Beyond, Calico **Sept.16.2017**

46. Differential stoichiometry among core ribosomal proteins
CSHL Translational Control, Cold Spring Harbor Laboratory **Sept.08.2016**

47. Principles of ribosome-mediated translational regulation
Center for Interdisciplinary Research on Complex Systems **Apr.12.2016**

48. Coordination among cell growth, cell division and gene expression
Broad Institute of MIT and Harvard, YSB **Dec.9.2015**

49. Coordinating cell growth and metabolism with ribosome-mediated translational regulation
Northeastern University, Biology Department Colloquium **Oct.19.2015**

50. Direct quantification of highly homologous proteins by mass–spectrometry indicates physiological changes in the protein composition of the ribosome
Broad Institute of MIT and Harvard, CBBO **Jun.25.2014**

51. Variable stoichiometry among core ribosomal proteins
Broad Institute of MIT and Harvard **Jun.19.2014**

52. Quantification of translational regulation of homologous proteins with unprecedented accuracy
Harvard HMS Systems Biology Theory Lunch **Mar.21.2014**

53. Exponentially growing cells can support a constant growth rate by very different metabolic strategies
Fourth Annual Physical Sciences in Oncology Meeting, NCI **Apr.18.2013**

54. Experimentally identified trade-offs of aerobic glycolysis
Center of Cancer Systems Biology at Tufts **Dec.18.2012**

55. Trade-offs of aerobic glycolysis
Harvard HMS Systems Biology Theory Lunch **Sep.12.2012**

56. The dynamics of exponential cell growth at constant growth rate
MIT Physical Sciences-Oncology Center **Jun.11.2012**

57. Metabolic cycling without cell division cycling
Bauer Forum at Harvard FAS Systems Biology **Apr.15.2011**

CONFERENCE TALKS

1. Quantifying homologous proteins and proteoforms
US HUPO Conference **Mar.14.2016**

2. Variable stoichiometry among core ribosomal proteins
2014 ASCB/IFCB Annual Meeting **Dec.8.2014**

3. The dynamics of exponential growth
GSA Yeast Meeting, Princeton University **Aug.5.2012**

4. The dynamics of exponential growth
Broad Institute of Harvard and MIT **Apr.17.2012**

5. Inference of Sparse Networks with Unobserved Variables. Application to Gene Regulatory Networks
International Conference on Artificial Intelligence and Statistics **May.14.2010**